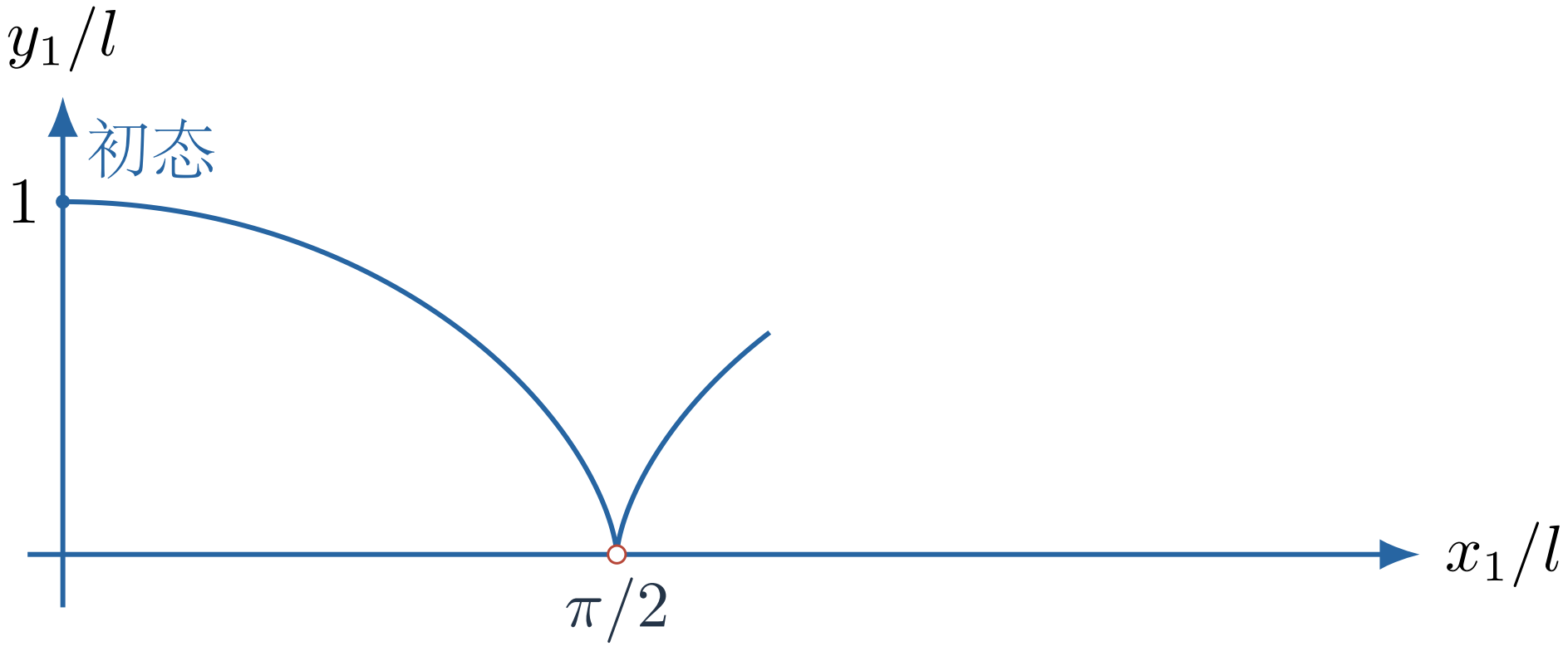

$\phi = \pi$： 瞬时零速